

From Ions to Chaos

The Larter–Breakspear model & a whole-brain framework in the mouse

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Part 2 — The Larter–Breakspear model and a recent publication

Workshop overview — how today is structured

Block	Content	Time
Part 1	The mouse, the data & the connectome	
Jeehye	Introduction to mouse data analysis	50 min
<i>Break</i>		<i>10 min</i>
Part 2	The Larter–Breakspear model	
Amaru	A recent publication: from ions to network-wide chaos	50 min
Q & A	joint, both presenters	10 min

What we will cover

- 1 The Larter–Breakspear model — what it is & why
- 2 A recent publication — four main findings:
 - reproduces mouse functional connectivity
 - chaos is **network-wide** (Lyapunov, Poincaré)
 - realistic delays reshape synchronization
 - ion channels set the regime; Ca gates stimulation

The thread of Part 2

How do we get from **ion channels** (Ca, Na, K) to **whole-brain network dynamics** we can compare against real mouse recordings?

ions → single population → chaos →
connectome → functional connectivity

The multi-scale problem

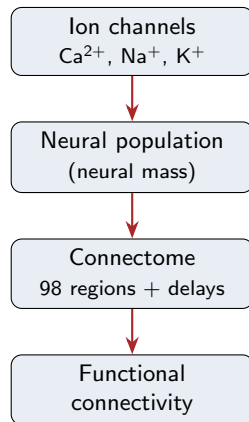
Computational neuroscience bridges scales:

- **Microscale:** ion channels, synapses — best measured in **mice** (knock-outs, opto-/electrophysiology)
- **Macroscale:** large-scale network dynamics, functional connectivity — measured with fMRI / LFP / EEG

The gap: rich micro-scale mouse data are rarely fed into *whole-brain* models.

Our goal

A model whose parameters *are* ion-channel conductances, embedded in a real mouse connectome and validated against mouse recordings.



What is a neural mass model?

- We do *not* simulate single spiking neurons.
- Instead: track the **mean activity** of a whole population (a “mass”) with a few differential equations.
- One node = one cortical region.

The Larter–Breakspear (LB) model is special:

- It is **conductance-based** — variables are real biophysical quantities (Ca, Na, K conductances; Nernst potentials).
- A smooth **sigmoidal input–output function** turns each population’s mean membrane potential into a mean firing rate (details in backup).
- It supports a rich repertoire: fixed points, limit cycles, and **deterministic chaos**. [Breakspear 2003]

The LB equations — two coupled state variables

Each node k has:

- V — mean **pyramidal membrane potential**
- W — fraction of open **K⁺ channels**
- Z — mean **inhibitory population potential**

Currents that shape $\dot{V}$:

- **Ca²⁺** (fast, excitatory)
- **Na⁺** (fast, excitatory)
- **K⁺** (slow, recovery → refractoriness)
- leak + inhibition Z

A **fast** (Ca/Na) and a **slow** (K, inhibition) timescale together generate the complex dynamics.

$$\begin{aligned}\dot{V} = & - (g_{Ca} + (1-C) r_{NMDA} a_{ee} Q_V \\ & + C r_{NMDA} a_{ee} \langle Q_V \rangle) m_{Ca} (V - V_{Ca}) \\ & - g_K W (V - V_K) - g_L (V - V_L) \\ & - (g_{Na} m_{Na} + (1-C) a_{ee} Q_V + C a_{ee} \langle Q_V \rangle) \\ & \times (V - V_{Na}) \\ & - a_{ie} Z Q_Z + a_{ne} I\end{aligned}$$

$$\dot{W} = \phi \frac{m_K - W}{\tau_K}$$

$$\dot{Z} = b(a_{ni} I + a_{ei} V Q_V)$$

$\langle Q_V \rangle^k$: long-range input from coupled nodes (C = coupling).
Sigmoidal firing rates Q_V, Q_Z and gating m_{ion} : see backup.

Why this model? A short genealogy

- **Larter et al. (1999); Breakspear et al. (2003)** — the model; chaos in a single cortical population.
- **Breakspear et al. (2003)** — coupling two/many populations: strong coupling → synchrony; moderate → clusters.
- **Heitmann & Breakspear (2018)** — added *uniform* conduction delays on a *binary* macaque connectome; delays of 6–10 ms produce synchronized clusters.
- **Chesebro et al. (2023); Lin et al. (2026)** — single-node bifurcation analyses: Neimark–Sacker, period-doubling, torus routes to chaos.

The gap we close

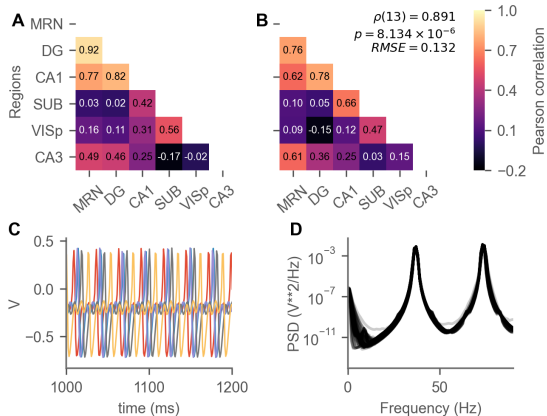
All prior network work used **binary connectomes** and/or **uniform delays**, and ion-channel analyses stayed at the **single node**. No one had run LB on a **realistic mouse connectome**.

A recent publication — the four contributions

- 1 **First integration** of LB into a tracer-derived **mouse connectome** (Allen Institute), validated against mouse LFP → reproduces empirical functional connectivity.
- 2 **Realistic connectome architecture matters:** empirical weights + **distance-dependent delays** qualitatively change synchronization.
- 3 **First whole-network characterization of chaos** in a biophysical model: Lyapunov exponents & Poincaré maps across all 98 populations.
- 4 **Ion channels at the network level:** Ca, Na, K each set global regimes; **calcium gates the network's response to stimulation.**

Munoz Lagunas, An, Stefanovski, Ritter — “From Ions to Chaos: a whole-brain modelling framework in the mouse” (submitted to *Comput. Methods Programs Biomed.*)

Result 1 — the model reproduces mouse functional connectivity



Setup

- 6 regions from Allen Institute mouse **LFP** recordings.
- Fit LB parameters by **stochastic grid search** to match the **functional-connectivity (FC)** matrix.

Outcome

- Empirical (A) vs. simulated (B) FC:
 $\rho = 0.89$, $p = 8 \times 10^{-6}$.
- (C) The fitted dynamics are the **chaotic, mixed-amplitude oscillations** typical of LB.
- (D) PSD: peaks at ~ 37 Hz (slow inhibitory rhythm) and ~ 73 Hz

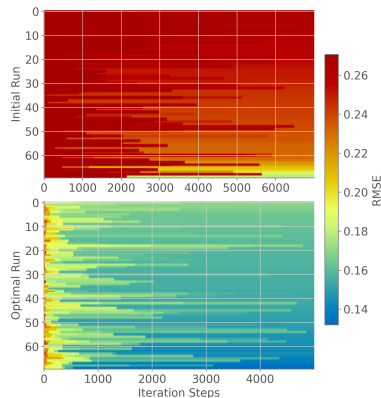
Result 1 (cont.) — how we fit the model on the HPC cluster

The fitting procedure (paper methods)

- Fit **9 model parameters** by a **stochastic grid search**, using the **RMSE** between simulated and empirical FC (6 Allen LFP regions) as the metric.
- **Two stages:**
 - *Coarse* — explore the global parameter space to localize the minimum region.
 - *Fine* — a narrowed search around that optimum: **70 parallel runs** $\times$ **5,000 combinations** each.

On the HPC cluster

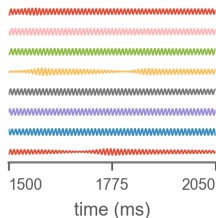
- Many fitting loops run **in parallel** on the BIH High-Performance Computing cluster.
- Best fit: **RMSE = 0.132**, $\rho = 0.89$ ($p = 8 \times 10^{-6}$).



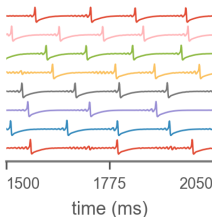
Suppl. Fig. S1 — RMSE trajectories across iterations; both stages converge to lower RMSE.

Result 2 — four dynamical regimes; δ_V as a tuning knob

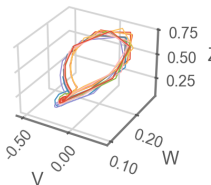
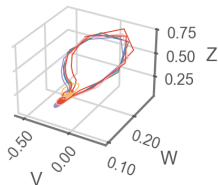
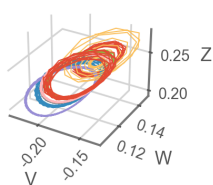
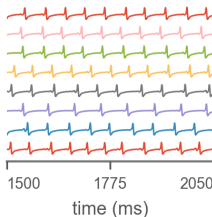
A $\delta_V = 0.53$



B $\delta_V = 0.55$



C $\delta_V = 0.68$



Varying the excitatory-threshold variance δ_V moves the network through:

- 1 **Fixed point**
- 2 **Fine-scale chaos** (fast, low amplitude)
- 3 **Mixed-amplitude chaos** (small + large excursions)
- 4 **Coexisting** regime

Top: excitatory time series. Bottom: phase planes. The transitions between small and large oscillations are **deterministic** — no noise required.

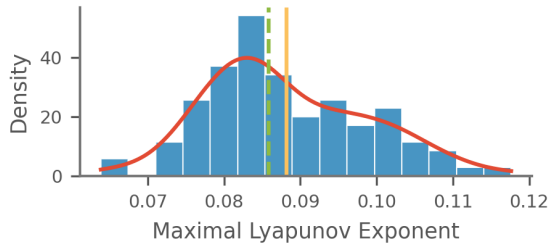
Result 3 — quantifying chaos I: Maximal Lyapunov Exponent (MLE)

Idea

- Take two nearby trajectories; measure how fast they **diverge**.
- Rate of divergence = Lyapunov exponent λ .
- $\lambda > 0 \Rightarrow$ **chaos** (exponential sensitivity to initial conditions).

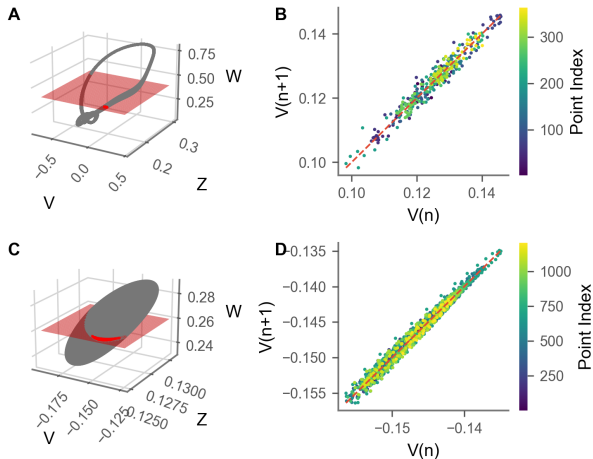
Result

- Computed λ for **all 98 populations**.
- Every node positive: mean $\bar{\lambda} = 0.088$ (Eckmann method), cross-validated 0.040 (Rosenstein).
- **Chaos is present network-wide.**



Distribution of MLEs across all cortical populations — all strictly positive.

Result 3 — quantifying chaos II: Poincaré maps



Idea

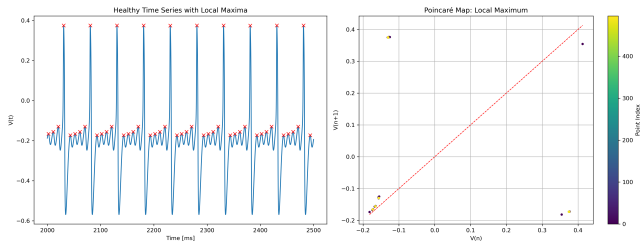
- Slice the 3-D phase space with a plane; record successive crossings $V(n) \rightarrow V(n+1)$.

Reading the map

- Periodic system $\rightarrow$ points collapse onto the **identity line** / a closed loop.
- Here: a **diffuse, non-repeating point cloud** that never settles — the signature of a **chaotic attractor**.

Two construction methods (plane crossing & local maxima) agree, and *distinguish* the chaotic regimes.

Result 3 — quantifying chaos II (cont.): the local-maximum method



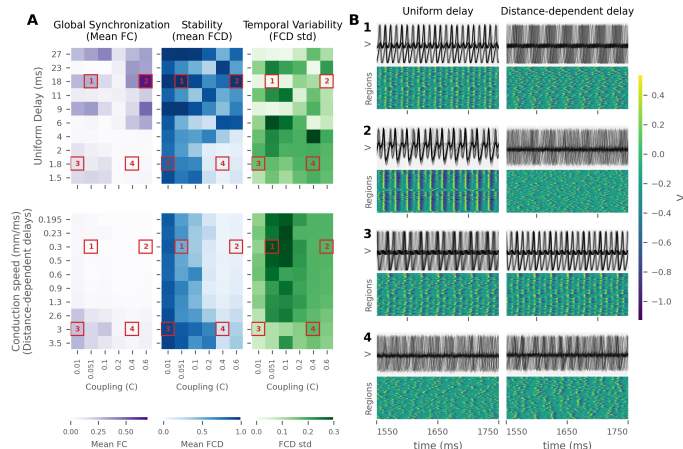
A plane-free construction

- Choosing a slicing plane is arbitrary; instead take the **successive local maxima** of $V(t)$ (red marks, left).
- Plot each peak against the next, $V_{\max}(n) \rightarrow V_{\max}(n+1)$ — a **first-return map** (right).

Reading the map

- Limit cycle $\rightarrow$ one point; period- $k \rightarrow k$ points.
- Here: **two amplitude bands** that never settle — **mixed-amplitude chaos**.
- Same verdict as the plane method, but robust and parameter-free.

Result 4 — realistic delays change synchronization



Compare two delay models, sweeping coupling & delay:

- Top: **constant** delays (as in prior work)
- Bottom: **distance-dependent** delays (realistic)

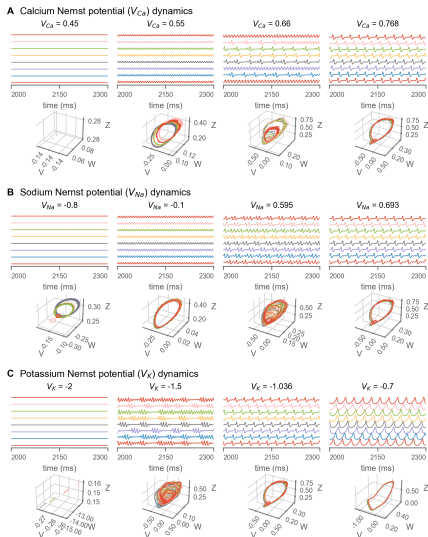
mean FC = **global synchrony**

mean FCD = **temporal stability**

FCD std = **variability**

Key point: matched *average* delay, very different dynamics. Constant delays at multiples of 9 ms give clusters (cf. Heitmann 2018); these *vanish* with realistic delays.

Result 5 — ion channels set the global regime

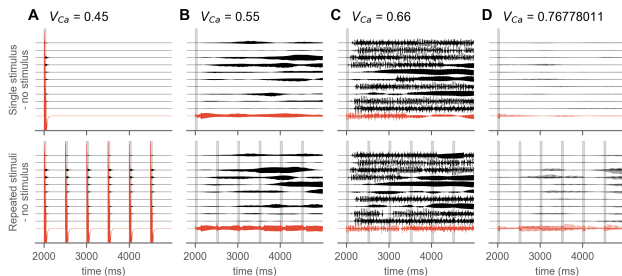


Sweeping the Nernst potentials at the **network** level (extending the single-node analysis of Chesebro 2023):

- V_{Ca} , V_{Na} , V_K each drive transitions fixed point $\rightarrow$ fine-scale $\rightarrow$ mixed-amplitude chaos.
- Low conductance $\rightarrow$ silence; high $\rightarrow$ sharp, high-amplitude spiking.

Top row: time series of 8 nodes; bottom: phase planes. Same machinery, three different ionic “control knobs”.

Result 5 (cont.) — calcium gates stimulus sensitivity

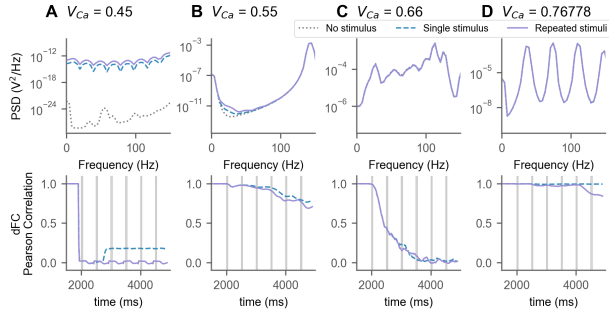


Plot **stimulus** — **no-stimulus**
(deterministic $\Rightarrow$ isolates the perturbation):

- **Fixed point** ($V_{Ca}=0.45$): transient only.
- **Fine-scale chaos**: a single pulse has lasting effects.
- **Coexisting**: **most sensitive** — one pulse reorganizes the whole network.
- **Mixed-amplitude**: robust to single pulses; repeated stimulation accumulates.

Calcium level $\Rightarrow$ how perturbable the network is.

Result 5 (cont.) — what stimulation actually changes



- Top: power spectra are **essentially unchanged** by stimulation (except the fixed-point regime, where stimulation “wakes up” silent nodes).
- Bottom: correlation of dynamic FC with vs. without stimulation **drops** — the more chaotic the regime, the more the *spatiotemporal pattern* is reorganized.

Take-home

Stimulation reshapes **where and when** regions co-activate (FC dynamics), not *which frequencies* they use.

What we built

- LB conductance-based model in the **Allen mouse connectome** with distance-dependent delays, on **TVB**.

What we found

- Reproduces mouse FC ($\rho = 0.89$).
- Chaos is **network-wide** (MLE & Poincaré, all 98 nodes).
- Realistic delays reshape synchronization.
- Ca/Na/K set regimes; **Ca gates stimulus sensitivity**.

Why it matters — what's next

We now have a **validated, biophysical, whole-brain mouse model** whose knobs are ion channels and whose inhibition we can manipulate.

Key references I

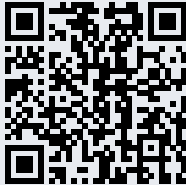
- [1] Larter R., Speelman B., Worth R.M. (1999). A coupled ordinary differential equation lattice model for the simulation of epileptic seizures. *Chaos* 9(3):795–804.
- [2] Breakspear M., Terry J.R., Friston K.J. (2003). Modulation of excitatory synaptic coupling facilitates synchronization and complex dynamics in a biophysical model of neuronal dynamics. *Network: Comput. Neural Syst.* 14:703–732.
- [3] Breakspear M., Terry J.R. (2005). Dynamics of a neural system with a multiscale architecture. *Neural Networks* 18:737–758.
- [4] Heitmann S., Breakspear M. (2018). Putting the “dynamic” back into dynamic functional connectivity. *Network Neuroscience* 2(2):150–174.
- [5] Chesebro A.G., Mujica-Parodi L.R., Weistuch C. (2023). Ion gradient-driven bifurcations of a multi-scale neuronal model. *Chaos, Solitons & Fractals* 167:113120.
- [6] Lin et al. (2026). Torus bifurcations and Shilnikov homoclinic orbits in the Larter–Breakspear model.
- [7] Oh S.W. et al. (2014). A mesoscale connectome of the mouse brain. *Nature* 508:207–214.

Key references II

- [8] Siegle J.H. et al. (2021). Survey of spiking in the mouse visual system reveals functional hierarchy. *Nature* 592:86–92.
- [9] Sanz-Leon P. et al. (2015). Mathematical framework for large-scale brain network modeling in The Virtual Brain. *NeuroImage* 111:385–430.
- [10] Ritter P. et al. (2013). The Virtual Brain integrates computational modeling and multimodal neuroimaging. *Brain Connectivity* 3(2):121–145.
- [11] Melozzi F. et al. (2017). The Virtual Mouse Brain. *eNeuro* 4(3).
- [12] Eckmann J.-P. et al. (1986). Liapunov exponents from time series. *Phys. Rev. A* 34:4971.
- [13] Rosenstein M.T. et al. (1993). A practical method for calculating largest Lyapunov exponents. *Physica D* 65:117–134.
- [14] Strogatz S.H. (1994). *Nonlinear Dynamics and Chaos*.

Thank you!

Questions welcome — or hold them for the joint Q & A.



Read the preprint — *From Ions to Chaos*

biorxiv.org/content/10.64898/2025.12.04.691835v1

Backup — the sigmoidal input–output functions

Each ionic gating fraction and each population firing rate is a **smooth sigmoid** of membrane potential u :

$$m_{\text{ion}}(u) = \frac{1}{2} \left(1 + \tanh \frac{u - T_{\text{ion}}}{\delta_{\text{ion}}} \right),$$
$$Q_V(u) = Q_V^{\max} \left(1 + \tanh \frac{u - V_T}{\delta_V} \right),$$
$$Q_Z(u) = Q_Z^{\max} \left(1 + \tanh \frac{u - Z_T}{\delta_Z} \right).$$

with $\text{ion} \in \{Ca, Na, K\}$.

- T_{ion} sets the **half-activation** potential; the channel opens as u rises past it.
- δ controls the **steepness**: small $\delta \rightarrow$ near step-like switching, large $\delta \rightarrow$ graded response.
- δ_V (excitatory threshold variance) is the main **tuning knob** that moves the network between fixed point, limit cycle and chaos.
- Because the population is a “mass” of many neurons with a spread of thresholds, the firing rate is graded rather than all-or-none — this is what makes the dynamics **continuous and differentiable**.